

PHYSICS

PAPER 1 (THEORY)

Maximum Marks: 70**Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 8 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given after these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1. Mass of an electron	m_e	=	$9.1 \times 10^{-31} \text{ kg}$
2. Charge of an electron (magnitude)	e	=	$1.6 \times 10^{-19} \text{ C}$
3. Planck's constant	h	=	$6.6 \times 10^{-34} \text{ J s}$
4. Speed of light in vacuum	c	=	$3 \times 10^8 \text{ m s}^{-1}$
5.	$1/4\pi\epsilon_0$	=	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
6. Permeability of vacuum	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
7. One electron volt	1 eV	=	$1.6 \times 10^{-19} \text{ J}$
8. Atomic mass unit	1 u	=	$931.5 \text{ MeV}/c^2 = 1.66 \times 10^{-27} \text{ kg}$
9. Rydberg constant	R	=	$1.097 \times 10^7 \text{ m}^{-1}$

Instruction to Supervising Examiner: Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS

Question 1

- (A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.
- (i) In a balanced Wheatstone bridge, the positions of the cell and the galvanometer are interchanged. The galvanometer will now: (*Understand*) [1]
- show a deflection in the same direction as before the bridge was balanced
 - show a deflection in the opposite direction
 - still show no deflection
 - show a deflection that depends on the emf of the cell
- (ii) Match the entries in Column I with those in Column II. [1]

Column I	Column II
1. Diamagnetic substance	A. χ small and positive, varying as $1/T$
2. Paramagnetic substance	B. χ small and negative, independent of temperature
3. Ferromagnetic substance	C. χ very large; becomes paramagnetic above the Curie temperature
4. Soft iron	D. low retentivity and low coercivity

(*Recall*)

- 1-A, 2-B, 3-C, 4-D
 - 1-B, 2-A, 3-D, 4-C
 - 1-B, 2-A, 3-C, 4-D
 - 1-C, 2-A, 3-B, 4-D
- (iii) Select the odd one out, on the basis of the principle on which they work. (*Analyse*) [1]
- AC generator
 - Transformer
 - Induction furnace
 - Electric room heater
- (iv) The slope of the graph of stopping potential V_s against the frequency ν of the incident radiation, for any photosensitive metal, is equal to: (*Recall*) [1]
- h
 - h/e
 - e/h
 - ϕ_0/e

- (v) Identify the correct statements about a reverse-biased p-n junction. [1]
- (P) The width of the depletion layer increases.
 (Q) The current is due mainly to the diffusion of majority carriers.
 (R) The height of the potential barrier increases.
 (S) The junction offers a very low resistance.
- (Understand)*
- (a) Both (P) and (Q)
 (b) Both (Q) and (S)
 (c) Both (R) and (S)
 (d) Both (P) and (R)
- (vi) An astronomical telescope has an objective of focal length 100 cm and an eyepiece of focal length 5 cm. In normal adjustment its magnifying power and length are: *(Apply)* [1]
- (a) 20 and 105 cm
 (b) 20 and 95 cm
 (c) 500 and 105 cm
 (d) 0.05 and 105 cm
- (vii) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]
- Assertion:** Diffraction of sound is noticed in everyday life far more easily than diffraction of light.
- Reason:** The wavelength of sound is comparable to the size of ordinary obstacles and openings, while that of light is very much smaller.
- (Analyse)*
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.
- (B) Answer the following questions briefly:
- (i) A point charge is enclosed at the centre of a spherical Gaussian surface. How does the electric flux through the surface change if its radius is doubled? *(Understand)* [1]
- (ii) A charged particle moves parallel to a uniform magnetic field. What is the magnetic force on it? *(Understand)* [1]
- (iii) Two thin lenses of focal lengths +20 cm and -40 cm are placed in contact. What is the power of the combination? *(Apply)* [1]
- (iv) Calculate the momentum of a photon of energy 4.8×10^{-19} J. *(Apply)* [1]
- (v) What is the function of the cadmium (or boron) rods in a nuclear reactor? *(Recall)* [1]
- (vi) What determines the colour of the light emitted by an LED? *(Recall)* [1]
- (vii) Fill in the blank with an appropriate term from the bracket. [1]
- In single-slit diffraction, the central maximum is _____ (half as wide as, as wide as, twice as wide as, four times as wide as) each secondary maximum. *(Recall)*

SECTION B – 14 MARKS

Question 2

- (i) Point charges of $+2\ \mu\text{C}$, $-3\ \mu\text{C}$ and $+4\ \mu\text{C}$ are held at the corners of an equilateral triangle of side 30 cm. Calculate the electrostatic potential energy of the system. (*Apply*) [2]

OR

- (ii) A parallel-plate capacitor is charged by a battery, which is then disconnected. A slab of dielectric constant K is now inserted to fill the space between the plates. How do (a) the capacitance, (b) the potential difference, (c) the electric field between the plates and (d) the stored energy change? (*Understand*) [2]

Question 3

- On open circuit, a cell is balanced against 110 cm of a potentiometer wire. When a resistance of $10\ \Omega$ is connected across the cell, the balancing length becomes 100 cm. Calculate the internal resistance of the cell. (*Apply*) [2]

Question 4

- (i) Figure 1 shows part of a circuit in which a current of 1 A flows from A to B. Calculate the potential difference $V_A - V_B$. (*Apply*) [2]

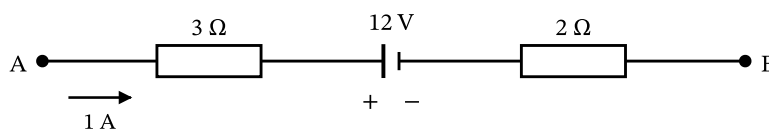


Figure 1

OR

- (ii) Two identical cells, each of emf E and internal resistance r , send the same current through an external resistance R whether they are connected in series or in parallel. Show that $R = r$. (*Analyse*) [2]

Question 5

- Two moving-coil galvanometers G_1 and G_2 have coils of equal area in the same radial field and with identical springs. G_1 has 50 turns and resistance $10\ \Omega$; G_2 has 20 turns and resistance $2\ \Omega$. Find the ratio of their (i) current sensitivities and (ii) voltage sensitivities. (*Apply*) [2]

Question 6

- (i) A thin equiconvex glass lens of focal length 20 cm is cut into two identical plano-convex halves by a plane perpendicular to its principal axis. What is the focal length of each half? (*Analyse*) [1]
- (ii) If the glass has refractive index 1.5, what is the power of one half when it is immersed in water (refractive index $4/3$)? (*Apply*) [1]

Question 7

- (i) Name the electromagnetic radiation used to scan luggage at airports and give its approximate wavelength range. (*Recall*) [1]

- (ii) A radio station broadcasts at 30 MHz. Calculate the wavelength of its waves. (*Apply*) [1]

Question 8

- A proton and an α -particle, both starting from rest, are accelerated through the same potential difference. Find the ratio of their de Broglie wavelengths. (*Analyse*) [2]

SECTION C – 27 MARKS

Question 9

- (i) Obtain an expression for the electric field intensity at a point on the equatorial line of an electric dipole of moment p , at a distance r from its centre. (*Understand*) [3]

OR

- (ii) Derive an expression for the electric potential at a distance r from a point charge Q . (*Understand*) [3]

Question 10

- (i) A potentiometer wire of length 4 m and resistance $10\ \Omega$ is connected in series with a resistance of $9\ \Omega$ and a driver cell of emf 4 V and internal resistance $1\ \Omega$. Calculate (a) the current in the wire, (b) the potential gradient along it, and (c) the emf of a cell that is balanced by 280 cm of the wire. (*Apply*) [3]

OR

- (ii) Twelve identical cells, each of emf 1.5 V and internal resistance $0.5\ \Omega$, are arranged in 3 parallel rows of 4 cells in series. The battery is connected to an external resistance of $1\ \Omega$. Calculate the current through the external resistance and the terminal voltage of the battery. (*Apply*) [3]

Question 11

- State Ampère's circuital law. Use it to obtain an expression for the magnetic field at a distance r from an infinitely long straight wire carrying a current I . (*Understand*) [3]

Question 12

- (i) An electron moving with a speed of $8 \times 10^6\ \text{m s}^{-1}$ enters a uniform magnetic field of $3 \times 10^{-3}\ \text{T}$ at right angles to it. Why does it move in a circle? Calculate the radius of its path. (*Apply*) [2]
- (ii) The north pole of a bar magnet is moved towards a coil. As seen from the magnet, in which direction does the induced current flow in the coil? Name the law used. (*Analyse*) [1]

Question 13

A physics teacher drew the arrangement of Young's double-slit experiment shown in Figure 2 and asked the following questions. State whether each student's answer is correct or incorrect. Correct an incorrect response with a brief explanation.

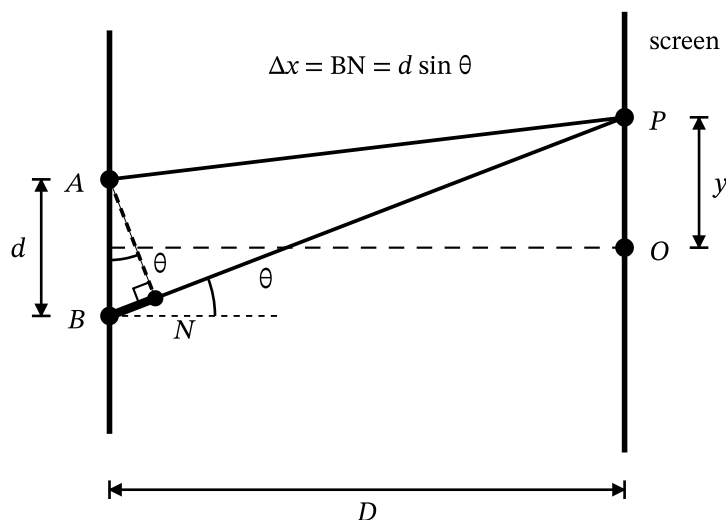


Figure 2

- (i) **Teacher:** At a point P on the screen the path difference is $1.5 \mu\text{m}$. If the wavelength is 600 nm , is there a bright or a dark fringe at P? [1]
Student: A bright fringe. (*Evaluate*)
- (ii) **Teacher:** If the slit separation is halved and the screen is moved to twice its distance, what happens to the fringe width? [1]
Student: It becomes four times as large. (*Evaluate*)
- (iii) **Teacher:** What will be seen if monochromatic light is replaced by white light? [1]
Student: All the fringes will be white. (*Evaluate*)

Question 14

Using Huygens' principle, draw a diagram showing the refraction of a plane wavefront at a plane surface separating a rarer medium from a denser one, and hence prove Snell's law. [3]
(*Understand*)

Question 15

- (i) A ray of light is incident normally on one face of a prism of angle 30° and refractive index $\sqrt{2}$. Calculate the angle of emergence and the angle of deviation. [2]
(*Apply*)
- (ii) Define the dispersive power of the material of a prism. [1]
(*Understand*)

Question 16

- (i) Draw a labelled ray diagram of a compound microscope forming the final image at the least distance of distinct vision. [1]
(*Understand*)
- (ii) The objective and eyepiece of a compound microscope have focal lengths 1.5 cm and 5 cm . An object is placed 1.6 cm from the objective and the final image is formed at 25 cm from the eyepiece. Calculate the magnifying power. [2]
(*Apply*)

Question 17

Light of wavelength 400 nm falls on two metal surfaces M_1 and M_2 whose work functions are 2.0 eV and 4.0 eV respectively.

- (i) Calculate the energy of a photon of this light in eV. (*Apply*) [1]
- (ii) Which surface emits photoelectrons? Give a reason. (*Analyse*) [1]
- (iii) Find the maximum kinetic energy of the emitted electrons and the stopping potential. (*Apply*) [1]

SECTION D – 15 MARKS**Question 18**

- (i) (a) Figure 3 shows how the emf $e = 200 \sin \omega t$ of an ac source and the current I through an electrical device connected to it vary with time.

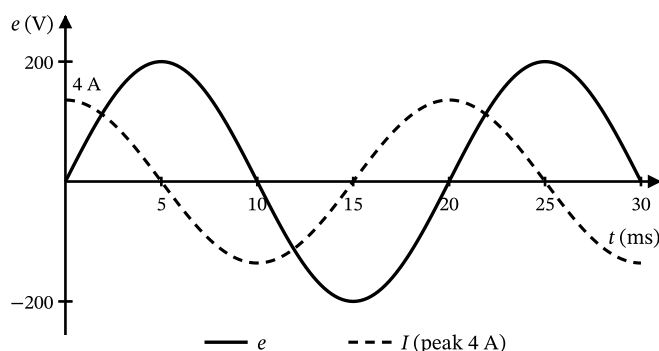


Figure 3

- (1) Identify the device (resistor, inductor or capacitor). Give a reason. (*Analyse*) [1]
- (2) Find its reactance. (*Apply*) [1]
- (3) Calculate its capacitance. (*Apply*) [1]
- (b) A metal rod slides along two parallel rails 0.5 m apart in a uniform vertical magnetic field of 0.4 T. Its position is $x = (2t^2 + 3)$ m. The rails are joined by a 4Ω resistor. Find the induced emf and current at $t = 2.5$ s. (*Apply*) [2]

OR

- (ii) (a) A coil draws a current of 2 A from a 100 V dc supply and 1 A from a 100 V, 50 Hz ac supply. Calculate its resistance and self-inductance. (*Apply*) [2]
- (b) What is the power factor of the coil on the ac supply? (*Apply*) [1]
- (c) Draw graphs showing how the reactances X_L and X_C vary with the frequency of the source. (*Recall*) [1]
- (d) What is meant by wattless current? (*Understand*) [1]

Question 19

- (i) (a) Using Bohr's postulates, obtain an expression for the total energy of the electron in the n th orbit of a hydrogen atom and show that $E_n \propto -1/n^2$. (*Apply*) [3]

- (b) Calculate the energy released in the fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He}$. Masses: ${}^2_1\text{H} = 2.014102 \text{ u}$, ${}^4_2\text{He} = 4.002604 \text{ u}$. (*Apply*) [2]

OR

- (ii) (a) Draw a graph of binding energy per nucleon against mass number. Mark the region in which nuclei release energy by fusion and that in which they release energy by fission. (*Understand*) [2]
- (b) Calculate the wavelength of the first line (H_α) of the Balmer series of hydrogen. (*Apply*) [2]
- (c) Why is the total energy of the electron in a hydrogen atom negative? (*Understand*) [1]

Question 20

Read the passage and answer the questions that follow.

Meera opens an old mobile-phone charger. A small transformer steps the 230 V mains down to about 9 V ac. This goes to a rectifier. She first finds a design with a single junction diode and a load, and then a better one that uses two diodes and a centre-tapped transformer. The rectified output is smoothed and kept steady at 5 V by a Zener diode, and a small red LED glows when the charger is switched on.

- (i) Draw the circuit of the single-diode rectifier and show its input and output waveforms. (*Understand*) [2]
- (ii) If the mains frequency is 50 Hz, what is the frequency of the output ripple of the single-diode design and of the two-diode design? (*Apply*) [1]
- (iii) In which bias is the Zener diode connected, and how does it keep the output at 5 V? (*Understand*) [1]
- (iv) Meera wants the indicator to glow blue instead of red. What property must the semiconductor of the new LED have? (*Analyse*) [1]

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